KARUSH SURI

Homepage ◇ Email ◇ GitHub ◇ Google Scholar ◇ LinkedIn ◇ X

AUTODIDACTIC LEARNING

Reading

Topology, James Munkres, all chapters	2026
Topics in Algebra, I.N. Herstein, all chapters	2025
Optimization Algorithms on Matrix Manifolds, P. Absil, chapters 2-7	2021
Convex Optimization, Stephen Boyd, chapters 6-11	2021
Neuro-dynamic Programming, Dimitri Bertsekas and John Tsitsiklis, chapters 1-6	2021
Abstract Dynamic Programming, Dimitri Bertsekas, chaters 1-2	2020
Machine Learning: A Probabilistic Perspective, Kevin Murphy, chapters 20-24	2020
Pattern Recognition and Machine Learning, Christopher Bishop, chapters 8-12	2019

Writing

Paper Summaries	(link)
Research Work	(link)

Code Implementations

Toy Examples (JAX and PyTorch)	(link)
Hierarchical DQN (JAX)	(link)
Proximal Policy Optimization (JAX)	(link)
Conservative Q Learning (JAX)	(link)
Discrete Gumbel Samplers (JAX)	(link)
Eligibility Traces with Neural Networks (PyTorch)	(link)
Evolution Strategies (PyTorch)	(link)
Lagrangian Algorithms and Vector Products (Autograd)	(link)

EDUCATION

University of Toronto 2025 - 2028 (Expected)
PhD, Electrical & Computer Engineering *Toronto, Canada*
Thesis: Tail End Gumbel Post-Training
GPA: 3.8 / 4 (coursework in mathematics)

University of Toronto 2019 - 2021
MASc, Electrical & Computer Engineering *Toronto, Canada*
Thesis: Deep Hierarchical Reinforcement Learning
GPA: 4/4

Amity University 2015 - 2019
BTech, Electrical & Computer Engineering *Delhi, India*
Thesis: Sign Language Translation from Wearable Sensors (demo)
GPA: 8.78/10 (rank: 4/142)

EMPLOYMENT

Valence Labs (Mila) 2023 - 2025
Research Engineer *Montreal, Canada*
Advisors: Dr. Emmanuel Bengio
Projects: **(1)** Multi-Modal Phenomic Foundational Models, **(2)** Amortized Inference with GFlowNets

Google X 2021 - 2023
AI Resident *Mountain View, USA*
Advisors: Grace Brentano & Dr. Lam Nguyen
Project: Undisclosed

INTERNSHIPS

Lossfunk 2026 - 2026
Research Intern *Toronto, Canada*
Advisors: Paras Chopra
Projects: Understanding Drug Repositioning via Chemical Reasoning

Fidelity Investments 2025 - 2026
Machine Learning Intern *Toronto, Canada*
Advisors: Kyle Burgers
Projects: Sparse Transformers for Tabular Quality Control

Borealis AI 2020 - 2021
Student Researcher *Toronto, Canada*
Advisors: Xiao Qi Shi
Projects: Deep Hierarchical Reinforcement Learning for Trade Execution

PUBLICATIONS

$*$ = equal contribution

"*Crosscoder Steering Structures and Stabilizes Multi-Answer Biological Reasoning*" Under Review
Karush Suri, Konstantinos Plataniotis, Yuri Lawryshyn

"*Understanding Judge Calibration in Multi-Turn Debates*" Under Review
Karush Suri, Konstantinos Plataniotis, Yuri Lawryshyn

"*Understanding and Mitigating Overconfidence in Focus Group Surveys*" TMLR 2026
Karush Suri, Konstantinos Plataniotis, Yuri Lawryshyn

"*A Cross Modal Knowledge Distillation & Data Augmentation Recipe for Improving Transcriptomics Representations through Morphological Features*" ICML 2025
Ihab Bendidi, Yassir El Mesbahi, Alisandra Kaye Denton, Karush Suri,
Kian Kenyon-Dean, Auguste Genovesio, Emmanuel Noutahi

"*Understanding Conditional Computation in Contrastive Phenomic Retrieval*" Technical Report 2025
Karush Suri, Puria Moghadam, Frederik Wenkel, Maciej Sypetkowski,
Emmanuel Bengio, Emmanuel Noutahi, Dominique Beaini

"*How Molecules Impact Cells: Unlocking Contrastive PhenoMolecular Retrieval*" NeurIPS 2024
Philip Fradkin*, Puria Moghadam*, Karush Suri, Frederik Wenkel,
Ali Bashashati, Maciej Sypetkowski, FM4S @ NeurIPS 2024 **(oral)**
Dominique Beaini **(best paper award)**

"*On the Scalability of GNNs for Molecular Graphs*" NeurIPS 2024
Maciej Sypetkowski, Frederik Wenkel, Farimah Poursafaei,
Nia Dickson, Karush Suri, Philip Fradkin, Dominique Beaini

"*Surprise Minimizing Multi-Agent Learning with Energy-based Models*" NeurIPS 2022
Karush Suri, Xiao Qi Shi, Konstantinos Plataniotis, Yuri Lawryshyn

"*Off-Policy Evolutionary Reinforcement Learning with Maximum Mutations*" AAMAS 2022 **(oral)**
Karush Suri

"*Continuous Sign Language Recognition from Wearable IMUs using Deep CapsNet and Game Theory*" CEE, Elsevier, 2019
Karush Suri, Rinki Gupta

"*Transfer Learning for sEMG-based Hand Gestures using Deep Learning in a Master- Slave Architecture*" IEEE IC3I 2018
Karush Suri, Rinki Gupta

PENDING PATENTS

"*Utilizing contrastive machine learning models to extract joint-space molecular-phenomic embeddings from molecular structures or phenomic images*" 2025
Dominique Beaini, Frederik Wenkel, Karush Suri,

Maciej Sypetkowski, Philip Fradkin, Puria Moghadam
Valence Labs, Application Number: WO2026096183A1.

"*Generating Actions for a Supply Chain Network*" 2024
Lam Nguyen, Grace Brentano, Sze Lee, Karush Suri, Anikait Singh,
Salil Pradhan, David Andre
Google X, Application Number: WO2025090795A1.

"*Large Language Model Interface for Supply Chain Networks*" 2023
Lam Nguyen, Grace Brentano, David Andre, Salil Pradhan, Anikait Singh, Karush Suri
Google X, Application Number: US20240289733A1.

SERVICES

Reviewer, *TMLR* 2026 - Present
Reviewer, *ICML* 2024 - 2026
Reviewer, *ICLR* 2024 - 2026
Reviewer, *NeurIPS* 2024 - 2026
Teaching Assistant, *Digital Image Processing*, University of Toronto 2021
Teaching Assistant, *Computational Thinking*, University of Toronto 2021
Teaching Assistant, *Computer Networks*, University of Toronto 2020
Teaching Assistant, *Computer Organization*, University of Toronto 2020

AWARDS

Academic

NSERC Canada Graduate Research Scholarship 2026 - 2029
University of Toronto Doctoral Scholarship 2025 - 2029
Hatch Graduate Scholarship 2025 - 2026
Laura Bassi Scholarship (declined) 2025
Electrical & Computer Engineering Fellowship, University of Toronto 2020 - 2021
Edward S. Rogers Graduate Scholarship, University of Toronto 2019 - 2020
Best in Technical Innovation Award (Class of 2015-2019), Amity University 2019
Most Frugal Innovation Award, Amity University 2018
100% Curriculum Merit Scholarship, Amity University 2015
Young Achievers in Mathematics Award 2015

Industrial

Outstanding Reviewer, NeurIPS, ICML 2023, 2026
Outstanding Intern Buddy, Google X 2022